

ECON 101 — Introduction to Microeconomics

Based on lectures by M. Skuterud

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These notes are a mathematical reorganisation of the lecture slides. The definitions and theorems are my own formalisation of the ideas in the slides; the examples and numbers come from the slides.

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1 The four core principles of economics

An *economy* is a system that answers four questions for a group of agents facing scarce resources: *what* to produce, *how* to produce it, *who* does what, and *who* gets what. (The slides' example is the island in Golding's *Lord of the Flies*.) The basic unit of analysis is an *individual decision*, and four principles tell us how decisions are made:

- (i) the cost–benefit principle,
- (ii) the opportunity cost principle,
- (iii) the marginal principle,
- (iv) the interdependence principle.

1.1 The cost–benefit principle

Throughout, an agent chooses from a set A of feasible options.

Definition. For an option $a \in A$, let $B(a)$ be its *benefit* and $C(a)$ its *cost*, both measured in dollars. The *economic surplus* of a is

$$S(a) = B(a) - C(a).$$

Benefits and costs need not be monetary. They are converted to dollars through willingness to pay.

Definition. The *willingness to pay* for a benefit is the largest amount of money w the agent would give up to get it. For a cost, it is the largest w the agent would pay to avoid it.

Principle (Cost–benefit principle). Pursue an option a if and only if $B(a) \geq C(a)$, i.e. $S(a) \geq 0$.

Remark. Some complications (the slides list them but do not resolve them yet):

- not everything is about money, which is why willingness to pay is used as the unit;
- the benefit may go to someone else;
- B and C may be uncertain, in which case they are random variables.

Costs and benefits are the *economic incentives* that drive decisions.

Gains from trade

Proposition 1.1 (Voluntary exchange is not zero-sum). A buyer values a good at v , a seller can produce it at cost c , and they trade at price p . Then

$$S_{\text{buyer}} = v - p, \quad S_{\text{seller}} = p - c, \quad S_{\text{buyer}} + S_{\text{seller}} = v - c.$$

Both agree to trade if and only if $c \leq p \leq v$, in which case both surpluses are nonnegative. The total surplus $v - c$ does not depend on p . The price only decides how it is split.

Proof. The buyer accepts iff $v - p \geq 0$ and the seller accepts iff $p - c \geq 0$. Adding the two surpluses cancels p . $\square$

Voluntary exchange creates surplus $v - c \geq 0$. Nobody loses.

1.2 The opportunity cost principle

Definition. Given a choice $a \in A$, the *next best alternative* is the best option in $A \setminus \{a\}$. The *opportunity cost* of a is the value of what is given up:

$$\text{OC}(a) = \max_{a' \in A \setminus \{a\}} (\text{value of } a').$$

The out-of-pocket part of a cost is the *financial cost*. The opportunity cost can be much larger.

Principle (Opportunity cost principle). The true cost of something is what you must give up to get it. When deciding on a , always ask “ a , or the next best alternative?”

Resources (income, time, energy) are *scarce*. That is why every choice has an opportunity cost.

Definition. A *sunk cost* is a cost that has to be paid whatever option is chosen, i.e. a cost k that enters $C(a)$ for every $a \in A$.

Proposition 1.2 (Ignore sunk costs). If $C(a) = \tilde{C}(a) + k$ for all $a \in A$ with k constant, then

$$\arg \max_{a \in A} (B(a) - C(a)) = \arg \max_{a \in A} (B(a) - \tilde{C}(a)).$$

Proof. Subtracting the constant k from every surplus preserves their order. $\square$

Example. Tuition is \$4,500 per term for 5 courses with 24 lectures each, so one lecture costs $\frac{4500}{5 \cdot 24} = \37.50 . You pay this whether you attend or not, so it is sunk and does not affect the decision. The relevant cost of going to lecture is the next best use of those 80 minutes, e.g. wages from a job.

Example. The biggest cost of an hour of gaming is not the hardware or software. It is the hour, which could have been spent on something else.

Exercise. Explain each of the following with the opportunity cost principle.

- (i) Many Canadians date as teenagers but few marry as teenagers.
- (ii) There are fewer stay-at-home mothers than 60 years ago.
- (iii) More Canadians are born in September–October than in February–March.
- (iv) Terminally ill patients are more willing to try unproven drugs.

Remark (Sketch). In (ii), higher wages for women raise the opportunity cost of staying at home. In (iv), the patient gives up fewer expected years of life, so the opportunity cost of the risk is lower.

Production possibilities frontier

Definition. Given scarce resources, the *feasible set* $F \subseteq \mathbb{R}_{\geq 0}^2$ is the set of output pairs (x, y) that can be produced. A point $(x, y) \in F$ is *efficient* if there is no $(x', y') \in F$ with $x' \geq x$, $y' \geq y$ and $(x', y') \neq (x, y)$. The *production possibilities frontier* (PPF) is the set of efficient points in F .

On a PPF $y = f(x)$ with f decreasing, the opportunity cost of one more unit of x is the amount of y given up, $-\Delta y / \Delta x$. For a differentiable f this is $|f'(x)|$, the absolute value of the slope.

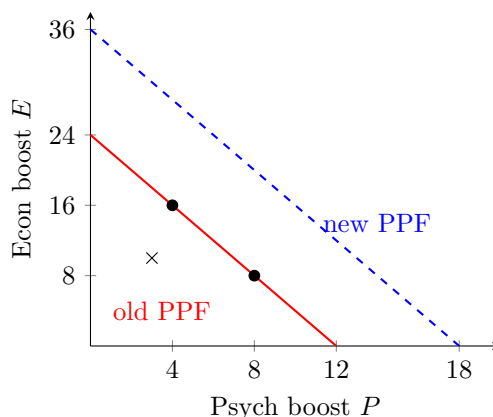
Example (Studying). You have 3 hours a night for economics (h_e) and psychology (h_p), so $h_e + h_p \leq 3$. Each hour raises your economics grade by 8 points and your psychology grade by 4 points:

$$E = 8h_e, \quad P = 4h_p.$$

Substituting $h_e = E/8$ and $h_p = P/4$ gives

$$F = \left\{ (P, E) \in \mathbb{R}_{\geq 0}^2 : \frac{E}{8} + \frac{P}{4} \leq 3 \right\}, \quad \text{PPF: } E = 24 - 2P, \quad 0 \leq P \leq 12.$$

The opportunity cost of one psychology point is 2 economics points. For example, going from (4, 16) to (8, 8) costs 8 economics points for 4 psychology points. Points strictly inside F , such as (3, 10), are *inefficient*.



Proposition 1.3 (Features of a PPF).

- (i) Every point on the PPF is efficient: one output can only be raised by lowering the other.
- (ii) The absolute value of the slope of the PPF is the opportunity cost of the good on the horizontal axis.
- (iii) Higher productivity enlarges F and pushes the PPF outward.

Example. In Example 1.2, suppose productivity rises by 50% in both subjects: $E = 12h_e$ and $P = 6h_p$. The new PPF is $E = 36 - 2P$ for $0 \leq P \leq 18$. The frontier moves out, but the opportunity cost is still $12/6 = 2$.

1.3 The marginal principle

So far choices were binary. Many choices are about *how much*: choose a quantity $q \in \{0, 1, 2, \dots\}$.

Definition. Let $TB(q)$ and $TC(q)$ be the *total benefit* and *total cost* of quantity q . For $q \geq 1$ the *marginal benefit* and *marginal cost* of the q -th unit are

$$MB(q) = TB(q) - TB(q-1), \quad MC(q) = TC(q) - TC(q-1).$$

The surplus is $S(q) = TB(q) - TC(q)$.

Principle (Marginal principle). Break a decision about quantity into a sequence of marginal decisions: “should I get one more?” Answer each one with the cost–benefit principle, i.e. compare $MB(q+1)$ with $MC(q+1)$.

Lemma 1.4. For all $q \geq 0$,

$$S(q) = S(0) + \sum_{k=1}^q (MB(k) - MC(k)).$$

Proof. The sum telescopes: $\sum_{k=1}^q MB(k) = TB(q) - TB(0)$, and similarly for MC . □

Theorem 1.5 (Rational rule). Suppose $MB - MC$ is nonincreasing (this holds, for instance, if MB is nonincreasing and MC is nondecreasing). Then

$$q^* = \max \{q \geq 0 : q = 0 \text{ or } MB(q) \geq MC(q)\}$$

maximises S . In words: *keep going as long as the marginal benefit is at least the marginal cost*.

Proof. Write $d(k) = MB(k) - MC(k)$, which is nonincreasing. By the choice of q^* and monotonicity, $d(k) \geq 0$ for $k \leq q^*$ and $d(k) < 0$ for $k > q^*$. By Lemma 1.4, if $q < q^*$ then $S(q^*) - S(q) = \sum_{k=q+1}^{q^*} d(k) \geq 0$, and if $q > q^*$ then $S(q) - S(q^*) = \sum_{k=q^*+1}^q d(k) < 0$. □

Remark. If q can vary continuously and TB, TC are differentiable, then $MB = TB'$ and $MC = TC'$. An interior optimum satisfies the first-order condition $S'(q^*) = 0$, i.e.

$$MB(q^*) = MC(q^*).$$

It is a maximum when $S'' = MB' - MC' \leq 0$. This is the “marginal benefit equals marginal cost” form of the rule.

Example (Nerida’s restaurant). Each meal sells for \$25. Total cost is \$10 per meal in food, \$300 per waiter, \$500 rent, and \$1,000 for Nerida’s own time. How many waiters should she hire?

Workers	Meals	TB	MB	TC	MC	S
2	160	4000	—	3700	—	300
3	210	5250	1250	4500	800	750
4	250	6250	1000	5200	700	1050
5	280	7000	750	5800	600	1200
6	300	7500	500	6300	500	1200
7	310	7750	250	6700	400	1050

For example, the 4th worker adds 40 meals, so $MB(4) = 25 \cdot 40 = 1000$ and $MC(4) = 300 + 10 \cdot 40 = 700$. Here both MB and MC fall, but $MB - MC = 450, 300, 150, 0, -150$ is still decreasing, so Theorem 1.5 applies. The rule stops at $q^* = 6$, where $MB = MC = 500$, and the maximum surplus is \$1,200. (The 6th worker adds exactly zero surplus, so $q = 5$ is also optimal.)

Remark. Rent and Nerida's time are the same for every q . They are sunk with respect to this decision and never appear in MC (Proposition 1.2).

1.4 The interdependence principle

Principle (Interdependence principle). Your best choice in one situation depends on:

- (i) *your other choices*: resources (income, time, attention, wealth) are shared across all your decisions;
- (ii) *other people's choices*: scarcity creates competition between agents (people, businesses, governments);
- (iii) *other markets*: e.g. labour, housing and credit markets affect one another;
- (iv) *time*: today's choice depends on past choices and expected future ones.

Remark. Mathematically, (i) says you solve one problem subject to a joint budget constraint (as in the PPF), not several problems separately. Points (ii) and (iii) say that the parameters of your problem, such as prices, are determined by the choices of everyone else.

2 Demand and consumer choice

2.1 Individual demand

Definition. The *individual demand curve* of a buyer i is the function

$$q_i^D : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}, \quad p \mapsto q_i^D(p),$$

giving the quantity demanded at each price p , *holding everything else constant* (*ceteris paribus*). By convention price goes on the vertical axis and quantity on the horizontal axis.

Example (Darren's demand for gasoline). A survey of Darren's daily gas use (litres) at hypothetical prices:

p (\$/litre)	2.40	2.20	2.00	1.80	1.60
$q_{\text{Darren}}^D(p)$	1	2	3	5	7

Law (Law of demand). Quantity demanded rises when the price falls, and vice versa:

$$p_1 < p_2 \implies q^D(p_1) \geq q^D(p_2).$$

That is, demand curves are *downward sloping*.

2.2 The rational rule for buyers

To explain the law of demand, apply the marginal principle to a buyer who takes the price p as given. Buying the k -th unit has marginal benefit $\text{MB}(k)$ and marginal cost p .

Theorem 2.1 (Rational rule for buyers). Suppose $\text{MB}(1) \geq \text{MB}(2) \geq \dots$ (*diminishing marginal benefit*). At price p the surplus-maximising quantity is

$$q^D(p) = \max \{k \geq 0 : k = 0 \text{ or } \text{MB}(k) \geq p\} = \# \{k \geq 1 : \text{MB}(k) \geq p\}.$$

At an interior optimum, price = marginal benefit.

Proof. Apply Theorem 1.5 with $\text{MC}(k) = p$ for all k . Since MB is nonincreasing, the set $\{k : \text{MB}(k) \geq p\}$ is an initial segment $\{1, \dots, q^D(p)\}$, so its largest element equals its size. $\square$

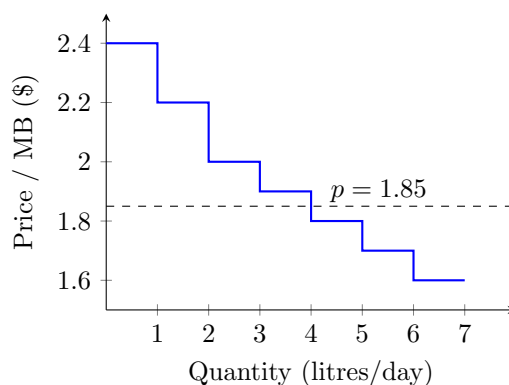
Corollary 2.2. Diminishing marginal benefit implies the law of demand. Moreover, the demand curve is the marginal benefit curve.

Proof. If $p_1 < p_2$ then $\{k : \text{MB}(k) \geq p_2\} \subseteq \{k : \text{MB}(k) \geq p_1\}$, so $q^D(p_2) \leq q^D(p_1)$ by Theorem 2.1. $\square$

Example. Darren ranks the uses of each extra litre:

k	Use of the k -th litre	$\text{MB}(k)$
1	drive to work instead of the bus	2.40
2	drive to school instead of biking	2.20
3	shop at the cheaper No Frills	2.00
4	drive to see friends	1.90
5	drive to the gym	1.80
6	drive to do errands	1.70
7	scenic drive	1.60

At $p = 1.85$, $\{k : \text{MB}(k) \geq 1.85\} = \{1, 2, 3, 4\}$, so Darren buys 4 litres a day. The formula also reproduces the survey in Example 2.1: e.g. at $p = 1.80$ he buys 5 litres.



2.3 Market demand

Definition. If the buyers are $i = 1, \dots, N$, the *market demand curve* is the horizontal sum of the individual demand curves:

$$Q^D(p) = \sum_{i=1}^N q_i^D(p).$$

Proposition 2.3. If every q_i^D is nonincreasing, then so is Q^D .

Proof. A sum of nonincreasing functions is nonincreasing. □

A movement along Q^D reflects two things: existing buyers changing the quantity they buy, and buyers entering or leaving the market ($q_i^D(p)$ changing between 0 and a positive value).

Example (Estimating market demand). Survey a random sample of $n = 400$ people from a population of $N = 40$ million. If $\bar{q}(p)$ is the average demand in the sample, then $Q^D(p) \approx N\bar{q}(p) = \frac{N}{n} \sum_{i \in \text{sample}} q_i^D(p)$, with scale factor $\frac{N}{n} = 100,000$.

p	Sample total (litres)	$\times N/n$	$Q^D(p)$ (millions)
1.60	1,200	10^5	120
1.80	1,100	10^5	110
2.00	1,000	10^5	100
2.20	900	10^5	90
2.40	800	10^5	80

2.4 Movements along versus shifts of demand

Demand really depends on many variables:

$$Q^D = Q^D(p; I, \theta, p_r, e, \dots, N),$$

where I is income, θ preferences, p_r prices of related goods, e expectations and N the number of buyers. The demand *curve* is the graph of $p \mapsto Q^D(p; \cdot)$ with all other arguments fixed.

Definition. A change in the good's own price p is a *movement along the demand curve* and changes the *quantity demanded*. A change in any other argument is a *shift of the demand curve* and changes *demand*. An *increase in demand* is a shift to the right (more demanded at every p). A *decrease* is a shift to the left.

Warning. “The price fell, so demand rose” is wrong. A price fall raises the *quantity demanded*; the curve itself does not move. For example, a price drop from \$2.20 to \$1.80 moves the market along the curve from 90 to 110 million litres.

The six demand shifters

(i) *Income.*

Definition. A good is *normal* if demand rises with income ($\partial Q^D / \partial I > 0$) and *inferior* if it falls ($\partial Q^D / \partial I < 0$).

(ii) *Preferences*, e.g. a marketing campaign convincing you that a brand is “cool”.

(iii) *Prices of related goods*.

Definition. Let p_y be the price of another good y . Then y is a *complement* of x if $\partial Q_x^D / \partial p_y < 0$ (a cheaper y raises demand for x), and a *substitute* if $\partial Q_x^D / \partial p_y > 0$ (a cheaper y lowers demand for x).

(iv) *Expectations*: beliefs about future prices affect demand today.

(v) *Congestion and network effects*.

Definition. There is a *network effect* if a good is worth more to each buyer when more other people use it, and a *congestion effect* if it is worth less.

(vi) *Type and number of buyers*. Since $Q^D = \sum_{i=1}^N q_i^D$, raising N shifts the *market* curve right. Unlike factors (i)–(v), it does not shift any individual curve q_i^D .

Example. A cut in income tax rates raises consumers’ disposable income I . If gas is a normal good, market demand for gas shifts to the right: at every price, more gas is demanded.

Exercise. Classify each change as a movement along or a shift of the demand for gasoline, and give the direction: (a) the price of gas falls; (b) the price of electric cars (a substitute for gas-powered driving) falls; (c) the population grows.

Remark (Answers). (a) Movement down along the curve, so quantity demanded rises. (b) Left shift, since the price of a substitute fell. (c) Right shift of the market curve (factor (vi)).

3 Supply and producer choice

This chapter mirrors Chapter 2 with sellers in place of buyers.

3.1 Individual supply

Definition. The *individual supply curve* of a seller j is the function $p \mapsto q_j^S(p)$ giving the quantity supplied at each price p , ceteris paribus.

Example (Shell). Shell's supply of gas (millions of litres per day):

$$q_{\text{Shell}}^S(p) = \begin{cases} 0 & p < 1.60, \\ 8 + 5(p - 1.60) & 1.60 \leq p \leq 2.40, \end{cases}$$

so $q^S(1.60) = 8$, $q^S(1.80) = 9$, $q^S(2.00) = 10$, $q^S(2.20) = 11$ and $q^S(2.40) = 12$. Below \$1.60 Shell stops supplying gas.

Law (Law of supply). Quantity supplied rises when the price rises, and vice versa:

$$p_1 < p_2 \implies q^S(p_1) \leq q^S(p_2).$$

That is, supply curves are *upward sloping*.

3.2 Perfect competition

Definition. A market is *perfectly competitive* if

- (i) there are many buyers and sellers, each small relative to the market, and no barriers to entry or exit;
- (ii) all firms sell an identical product;
- (iii) there is perfect information;
- (iv) mobility costs are zero: buyers can switch sellers for free.

An agent is a *price taker* if it treats the market price p as given, as opposed to a *price maker*.

Proposition 3.1. In a perfectly competitive market every firm sells at the market price p .

Argument. If a firm charges $p' > p$, then by (ii)–(iv) every buyer switches to a rival at no cost, so it sells nothing. Suppose instead that selling at some $p' < p$ were profitable. With perfect information and free entry, a competitor would already have taken that opportunity, so this cannot hold in a situation where nobody wants to change what they do. Hence p is exogenous to each firm. $\square$

Remark. Perfectly competitive markets are rare in reality. The assumption is a simplification that makes the analysis tractable.

3.3 The rational rule for sellers

Definition. A firm's total cost splits as

$$TC(q) = FC + VC(q),$$

where the *fixed cost* FC does not depend on output (e.g. capital) and the *variable cost* $VC(q)$ does (e.g. labour). The *marginal cost* is $MC(q) = TC(q) - TC(q-1)$, or $TC'(q)$ in the continuous case.

Proposition 3.2. $MC(q) = VC(q) - VC(q-1)$. In particular, fixed costs do not affect the choice of quantity.

Proof. FC cancels in the difference. This is Proposition 1.2 again. $\square$

For a price taker, the marginal benefit of selling one more unit is the price p .

Theorem 3.3 (Rational rule for sellers). If MC is nondecreasing, the profit-maximising quantity at price p is

$$q^S(p) = \max \{q \geq 0 : q = 0 \text{ or } MC(q) \leq p\}.$$

At an interior optimum, price = marginal cost. So: *sell one more unit if $p \geq MC$.*

Proof. Apply Theorem 1.5 with $MB(q) = p$ constant. $\square$

Corollary 3.4. Increasing marginal cost implies the law of supply, and the supply curve *is* the marginal cost curve.

Proof. If $p_1 < p_2$ then $\{q : MC(q) \leq p_1\} \subseteq \{q : MC(q) \leq p_2\}$. $\square$

Remark. The law of supply comes from two sources: *diminishing marginal product* (below) and *rising input costs*.

3.4 Diminishing marginal product

Definition. A *production function* $Y = F(L, K)$ gives output from labour L (variable input) and capital K (fixed input). The *marginal product of labour* is $MP_L = \partial F / \partial L$, or $F(L, K) - F(L - 1, K)$ in discrete terms. There is *diminishing marginal product* if MP_L decreases in L for fixed K .

Theorem 3.5 (Diminishing marginal product $\Rightarrow$ rising marginal cost). Let the wage be w and K fixed, and let $F(\cdot, K)$ be smooth and strictly increasing, with inverse $L(Y)$ (the labour needed to make Y). Then

$$MC(Y) = \frac{w}{MP_L(L(Y))}.$$

Hence if MP_L is decreasing, MC is increasing.

Proof. $TC(Y) = wL(Y) + FC$, so by the inverse function rule $MC(Y) = wL'(Y) = w/F_L(L(Y), K)$. When Y increases, $L(Y)$ increases, so MP_L falls and MC rises. $\square$

Each extra unit of output needs more and more labour, so each unit costs more than the last.

Example. Take $Y = \sqrt{LK}$ with $K = 4$, so $Y = 2\sqrt{L}$ and $MP_L = \frac{1}{2}\sqrt{K/L} = 1/\sqrt{L}$, which is decreasing. In discrete terms:

L	0	1	2	3	4	5	6
Y	0	2	2.83	3.46	4.00	4.47	4.90
MP_L	—	2	0.83	0.64	0.54	0.47	0.43

Invert: $L = Y^2/K = Y^2/4$. With $w = \$15$, $TC(Y) = 15 \cdot Y^2/4 = 3.75 Y^2$ (ignoring the fixed cost of capital), so

$$MC(Y) = TC(Y) - TC(Y - 1) = 3.75(2Y - 1).$$

Y	0	1	2	3	4	5	6
L	0	0.25	1	2.25	4	6.25	9
TC	0	3.75	15	33.75	60	93.75	135
MC	—	3.75	11.25	18.75	26.25	33.75	41.25

In the continuous version, $MC(Y) = TC'(Y) = 7.5 Y$, which matches Theorem 3.5: $w/MP_L(L(Y)) = 15\sqrt{Y^2/4} = 7.5 Y$. Setting $p = MC$ gives the supply curve $q^S(p) = p/7.5$, which is upward sloping.

Remark. The table's L column assumes labour can be hired in fractions. MP_L and MC move in opposite directions, as Theorem 3.5 predicts.

3.5 Market supply

Definition. The *market supply curve* is the horizontal sum $Q^S(p) = \sum_{j=1}^M q_j^S(p)$ over the M sellers. If the sellers are identical, $Q^S(p) = M q^S(p)$.

Example. With $M = 10$ identical firms:

p	1.60	1.80	2.00	2.20	2.40
$q^S(p)$	6	8	10	12	14
$Q^S(p) = 10 q^S(p)$	60	80	100	120	140

(millions of litres per day).

A higher price raises market quantity supplied in two ways: existing firms expand output, and new firms enter.

3.6 Movements along versus shifts of supply

As with demand, write

$$Q^S = Q^S(p; w, A, p_{\text{rel}}, e, M),$$

where w is input prices, A productivity, p_{rel} prices of related outputs, e expectations and M the number of sellers. A change in p is a *movement along the supply curve* and changes the *quantity supplied*. A change in any other argument is a *shift of the supply curve*. An increase in supply is a shift to the right.

The five supply shifters

- (i) *Input prices.* Higher input prices raise MC at every q , so supply shifts left.
- (ii) *Productivity and technology.* Producing more with the same inputs (new machinery, better management or processes, learning by doing) lowers MC, so supply shifts right.
- (iii) *Prices of related outputs.*

Definition. Good y is a *substitute-in-production* for x if it is made with the same inputs, and a *complement-in-production* if it is made together with x . Then

$$\frac{\partial Q_x^S}{\partial p_y} < 0 \text{ for substitutes,} \quad \frac{\partial Q_x^S}{\partial p_y} > 0 \text{ for complements.}$$

- (iv) *Expectations.* If firms expect the price to rise, they store output rather than sell it now, so supply today shifts left. This matters more for non-perishable goods (gasoline) than perishable ones (fresh fish).
- (v) *Type and number of sellers.* Changing M shifts the market curve but no individual curve. For example, a Competition Bureau ruling on a merger changes M .

Example. Crude oil is an input to gasoline. If its price rises, MC rises at every output level, so the market supply of gasoline shifts *left*: at every price, less is supplied. On the slides, supply at \$2.00 falls from 100 to 60 million litres.

Summary: buyers versus sellers

	Buyer	Seller
Marginal benefit	MB(q)	p
Marginal cost	p	MC(q)
Rational rule	$p = \text{MB}$	$p = \text{MC}$
Law	demand slopes down	supply slopes up
Reason	diminishing MB	diminishing MP_L , rising input costs
Market curve	$\sum_i q_i^D$	$\sum_j q_j^S$

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